

Swapping of two 8 bit dataAIM:

To write an assembly language to swap two 8 bit data using 8085 processor.

ALGORITHM:

- 1) Load the contents of memory address 1100 into A.
- 2) Move the contents of A to B.
- 3) Load the contents of 1101 into accumulator A.
- 4) Move contents of A into C.
- 5) Store contents of A into 1102.
- 6) Move content of register B to A.
- 7) Store content of A into memory address 1103.

PROGRAM:

```

LDA 1100 : load content of 1100 to A.
MOV B, A : Move contents of A to B
LDA 1101 : load content of 1101
           into A
MOV C, A : Move contents of A to C
STA 1102 : Store contents of A into

```

MOV A, B : Move contents of B to A

STA 1103 : store contents of A to 1103.

HLT : halt the program execution.

INPUT:

ADDRESS	DATA
1100	6
1101	4

OUTPUT:

ADDRESS	DATA
1103	4
1104	6

RESULT:

Thus the program was executed successfully using 8085 processor simulator.



Registers

A	06	S	0
BC	06 04	Z	0
DE	00 00	AC	0
HL	00 00	P	0
PSW	00 00	C	0
PC	42 10		
SP	FF FF		
Int-Reg	00		

Flag

Load me at

```
1 LDA 1100
2 MOV B, A
3 LDA 1101
4 MOV C, A
5 STA 1102
6 MOV A, B
7 STA 1103
8 HLT
```

Decimal - Hex Conversion

Decimal Hex

0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

1101	-	+	04
Update Memory			

Data Stack KeyPad Memory I/O Ports

Start 1102

OK

Address (Hex) Address Data

044E	1102	4
044F	1103	6
0450	1104	0
0451	1105	0
0452	1106	0
0453	1107	0
0454	1108	0
0455	1109	0
0456	1110	0
0457	1111	0
0458	1112	0
0459	1113	0
045A	1114	0
045B	1115	0
045C	1116	0
045D	1117	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

